## Layer 1 - Data Preprocessing: inputs and verified knowledge

**Raw Data Sources**
catalogue topic or paid learner request
primary history sources (web)
founder rules in the AvirmathAI skill

**Ingestion and Cleaning**
lesson schema validation (jsonschema)
source and date cross-check
placeholder binding to verified facts

**Mathematics as Code**
mathcore module (NumPy, SymPy)
facts dictionary: the only quotable numbers
unit tests bound to every equation

verified facts and lesson inputs

## Layer 2 - Orchestration and Reasoning

**Lesson Director (Orchestrator)**
job state and routing
goal persistence and run trace
bounded retries (max 3)

**Eight Specialised Agent Roles**
Profiler, Historian, Math Author,
Math Verifier, Lesson Writer,
Animation Director, Visual Critic,
Experiment Builder (plan, act, check)

**File-based Memory (no vector database)**
procedural: skill files
working: shared artifact store
long-term: project memory across sessions

agent requests: text, speech, frames

## Layer 3 - Core Models and Compute

**Language Model**
frontier LLM via an agent runtime
one model, all eight roles
image input for frame review

**Speech Model**
Microsoft neural TTS, en-GB Sonia
via edge-tts, word-level timings

**Deterministic Renderers and Compute**
Manim CE 0.21 + LaTeX frames
FFmpeg assembly, Python 3.13 site generator
local workstation, batched rendering

candidate artifacts

## Layer 4 - Governance and Execution Loop

**Tool Integration**
pytest and SymPy checks
schema validator
frame review of every rendered scene
browser-based page checks

**Verification Gates and Feedback**
failed gate returns the exact error
to the responsible agent
strategy recalibration, then retry

**Real-time Evaluation Metrics**
125 math tests passing
beat overrun = 0 s
frame defects = 0
facts coverage = 100%

gate results and metrics

## Layer 5 - Downstream Output and Safety

**Safety and Guardrail Boundaries**
numbers only from the facts dictionary
history only with sources
fixed branding and voice rules

**Observability and Human-in-the-Loop**
storyboard approval by the founder
run trace and review record
stop after 3 failed retries

**Downstream Delivery**
narrated video, captions, chapters
lesson web page with assessment
interactive experiment, SEO/AEO/GEO pack
static site deployment

feedback loops: a failed gate returns the error to the agents (Layer 4 to 2); learner feedback and on-demand requests return to the inputs (Layer 5 to 1)